Notebook 3 — Chapter 3.1

Sampling Mechanics & Finite Population Selection

20 worked examples — Hypergeometric models, Pólya urns, capture-recapture — calibrated for Stage 1 speed and Stage 2 trading intuition

A. Hypergeometric PMF — Basic Setup

Example 1 — The Basic Draw

An options desk's trade blotter has $N = 30$ fills, $K = 12$ marked "aggressive" (taker). You sample $n = 5$ fills without replacement. Find $\mathbb{P}$(exactly 3 aggressive).

Solution.

$$\mathbb{P}(X = 3) = \frac{\binom{12}{3}\binom{18}{2}}{\binom{30}{5}} = \frac{220 \cdot 153}{142506} = \boxed{0.2362}$$

⋆ *Intuition:* the numerator splits your choice into two independent sub-choices (which aggressive fills, which passive fills); the denominator is the total unrestricted way to pick 5 from 30. This split-and-normalize pattern is the entire hypergeometric machinery.

Example 2 — Complementary Counting

A lot of $N = 50$ chips has $K = 4$ defective. QA samples $n = 10$ without replacement. Find $\mathbb{P}$(at least 1 defective).

Solution. Complement is faster:

$$\mathbb{P}(X = 0) = \frac{\binom{4}{0}\binom{46}{10}}{\binom{50}{10}} = \frac{\binom{46}{10}}{\binom{50}{10}}$$

$$= \frac{46!/(10!36!)}{50!/(10!40!)} = \frac{40 \cdot 39 \cdot 38 \cdot 37}{50 \cdot 49 \cdot 48 \cdot 47} \approx 0.5125$$

$$\mathbb{P}(X \geq 1) = 1 - 0.5125 = \boxed{0.4875}$$

▶ *Speed trick:* for "at least 1" questions, write the ratio of falling factorials directly — $\prod_{i=0}^{n-1} \frac{N-K-i}{N-i}$ — and skip computing two huge binomial coefficients separately.

Example 3 — Hypergeometric vs. Binomial, Side by Side

$N = 10$ market makers, $K = 3$ are "toxic" flow sources. You interact with $n = 2$ of them. Compare $\mathbb{P}$(both toxic) under (a) sampling without replacement (Hypergeometric) and (b) sampling with replacement (Binomial, $p = 3/10$).

Solution. (a) $\mathbb{P} = \frac{\binom{3}{2}\binom{7}{0}}{\binom{10}{2}} = \frac{3}{45} = \boxed{0.0667}$. (b) $\mathbb{P} = p^2 = (0.3)^2 = \boxed{0.09}$.

⋆ *Intuition:* without replacement, drawing one toxic counterparty *removes* a toxic counterparty from the pool — the second draw is systematically less likely to repeat. This negative dependence is exactly why Hypergeometric variance is always $\leq$ Binomial variance for the same n, p (see Example 7).

Example 4 — Multivariate Hypergeometric

A book of $N = 20$ positions splits into 3 asset classes: 8 equities, 7 bonds, 5 derivatives. A risk audit samples $n = 6$ positions without replacement. Find $\mathbb{P}$(3 equities, 2 bonds, 1 derivative).

Solution.

$$\mathbb{P} = \frac{\binom{8}{3}\binom{7}{2}\binom{5}{1}}{\binom{20}{6}} = \frac{56 \cdot 21 \cdot 5}{38760} = \frac{5880}{38760} = \boxed{0.1517}$$

▶ *Generalization:* with c categories of sizes $K_1, \ldots, K_c$ ($\sum K_i = N$) and $\sum n_i = n$ drawn, $\mathbb{P} = \dfrac{\prod_i \binom{K_i}{n_i}}{\binom{N}{n}}$ — same split-and-normalize logic, just more factors on top.

B. Mean, Variance & the FPC

Example 5 — Mean via Indicators (No PMF Needed)

Derive $\mathbb{E}[X] = n\frac{K}{N}$ for Hypergeometric(N, K, n) without summing the PMF.

Solution. Let $I_j = 1$ if the j-th *marked* item is among the n drawn, $j = 1, \ldots, K$. By symmetry every marked item is equally likely to be in the sample: $\mathbb{P}(I_j = 1) = \frac{n}{N}$ (each of the N items is drawn with the same marginal probability n/N). Then $X = \sum_{j=1}^{K} I_j$, so

$$\mathbb{E}[X] = \sum_{j=1}^{K} \mathbb{E}[I_j] = K \cdot \frac{n}{N} = \boxed{\frac{nK}{N}}$$

⋆ *Speed trick:* this is identical in spirit to the indicator trick for $\mathbb{E}$[inversions] or $\mathbb{E}$[matches] — never integrate/sum a nasty PMF when linearity of expectation over 0/1 indicators gets there in two lines.

Example 6 — Variance via Indicator Covariance

Derive $\mathrm{Var}(X) = n\frac{K}{N}\left(1 - \frac{K}{N}\right)\frac{N-n}{N-1}$.

Solution. With I_j as in Example 5, $\mathrm{Var}(X) = \sum_j \mathrm{Var}(I_j) + \sum_{j \neq k} \mathrm{Cov}(I_j, I_k)$. $\mathrm{Var}(I_j) = \frac{n}{N}\left(1 - \frac{n}{N}\right)$. For $j \neq k$: $\mathbb{P}(I_j = I_k = 1) = \frac{n(n-1)}{N(N-1)}$, so

$$\mathrm{Cov}(I_j, I_k) = \frac{n(n-1)}{N(N-1)} - \frac{n^2}{N^2} = -\frac{n(N-n)}{N^2(N-1)}$$

Summing K variance terms and $K(K-1)$ covariance terms and simplifying:

$$\mathrm{Var}(X) = n \cdot \frac{K}{N} \cdot \frac{N-K}{N} \cdot \boxed{\frac{N-n}{N-1}}$$

▶ The last factor is the **finite population correction (FPC)**. It's negative-covariance bookkeeping, fully explicit.

Example 7 — FPC $\to 1$ as $N \to \infty$: Recovering Binomial

Show that as $N \to \infty$ with $K/N = p$ fixed and n fixed, Hypergeometric variance $\to$ Binomial variance $np(1-p)$.

Solution.

$$\frac{N-n}{N-1} = \frac{1 - n/N}{1 - 1/N} \xrightarrow{N \to \infty} 1$$

so $\mathrm{Var}(X) \to np(1-p)$, exactly Binomial. Also directly: as $N \to \infty$, removing one item barely changes the remaining proportion, so draws become effectively independent.

⋆ *Trading rule of thumb:* if your sample is $< 5\%$ of the population ($n/N < 0.05$), the FPC is within 5% of 1 and Binomial is a safe, simpler approximation. Audit samples from a 10,000-trade blotter can usually skip the FPC; a 30-trade universe cannot.

Example 8 — Variance Reduction from Sampling Without Replacement

A compliance team audits $n = 20$ trades from a population of $N = 100$, of which $K = 25$ breach a limit. Compute $\mathrm{Var}(X)$ exactly, and compare to the "naive" Binomial variance $np(1-p)$ that ignores the FPC.

Solution. $p = K/N = 0.25$. Naive: $np(1-p) = 20(0.25)(0.75) = 3.75$. FPC $= \frac{N-n}{N-1} = \frac{80}{99} = 0.8081$.

$$\mathrm{Var}(X) = 3.75 \times 0.8081 = \boxed{3.030}$$

▶ *Intuition:* ignoring the FPC here overstates variance (and thus a confidence interval's width) by about 24% — a real cost if you're sizing an audit sample to hit a target precision (see the formula sheet's sample-size inversion).

C. Approximations & Dualities

Example 9 — Binomial Approximation, Quantified

$N = 500$, $K = 100$ ($p = 0.2$), $n = 10$. Compare exact $\mathbb{P}(X = 2)$ to the Binomial$(10, 0.2)$ approximation.

Solution. Exact: $\mathbb{P}(X = 2) = \frac{\binom{100}{2}\binom{400}{8}}{\binom{500}{10}} = \boxed{0.3034}$ (to 4 d.p.). Binomial approx: $\binom{10}{2}(0.2)^2(0.8)^8 = 45(0.04)(0.1678) = \boxed{0.3020}$.

⋆ *Intuition:* agreement to 3 significant figures because $n/N = 2\% \ll 5\%$. The rule from Example 7 isn't just qualitative — it's this accurate in practice, and it's the single fastest legal shortcut on a Stage 1 MCQ with large N.

Example 10 — Normal Approximation with Continuity Correction

$N = 1000$, $K = 400$, $n = 100$. Approximate $\mathbb{P}(X \leq 30)$.

Solution. $\mu = n\frac{K}{N} = 40$. $\sigma^2 = n\frac{K}{N}\frac{N-K}{N}\frac{N-n}{N-1} = 100(0.4)(0.6)\frac{900}{999} = 21.62$, $\sigma \approx 4.650$. Continuity correction:

$$\mathbb{P}(X \leq 30) \approx \Phi\left(\frac{30.5 - 40}{4.650}\right) = \Phi(-2.043) \approx \boxed{0.0205}$$

▶ Same CLT machinery as Ch. 2.2, just with the Hypergeometric μ, σ^2 (FPC included) plugged in — always include the FPC in σ here, since $n/N = 10\%$ is not negligible.

Example 11 — The Symmetry Duality (Draws $\leftrightarrow$ Labels)

Show $\mathbb{P}(X = k)$ is unchanged if you instead fix the n draw *positions* and ask how many of the K marked items land in them, i.e. Hypergeometric$(N, K, n) \overset{d}{=}$ Hypergeometric(N, n, K).

Solution. Both count "size of overlap between a random n-subset and a

fixed K-subset of an N-set":

$$\frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}} = \frac{\binom{n}{k}\binom{N-n}{K-k}}{\binom{N}{K}}$$

(both sides count the same object: k marked items chosen for the draw, equivalently k draw-positions chosen for the marks.)
⋆ *Speed trick:* whichever of K or n is smaller, relabel so it plays the "choose k from" role — this can shrink a binomial coefficient from $\binom{500}{47}$ to $\binom{10}{3}$-sized arithmetic with zero loss of correctness.

D. Pólya Urn Schemes

Example 12 — Setup: the Reinforcing Draw

An urn starts with $r = 3$ red and $b = 2$ black balls. Draw a ball, note its color, return it *plus one more of the same color*. Find $\mathbb{P}$(2nd draw is red).
Solution. Law of total probability on the 1st draw:

$$\mathbb{P}(R_2) = \mathbb{P}(R_1)\mathbb{P}(R_2 \mid R_1) + \mathbb{P}(B_1)\mathbb{P}(R_2 \mid B_1)$$

$$= \frac{3}{5}\cdot\frac{4}{6} + \frac{2}{5}\cdot\frac{3}{6} = \frac{12}{30} + \frac{6}{30} = \frac{18}{30} = \boxed{\frac{3}{5}}$$

⋆ *Intuition:* identical to the initial proportion $\frac{r}{r+b} = \frac{3}{5}$! This is not a coincidence — it previews Example 13's exchangeability theorem: *every* draw in a Pólya urn has the same marginal distribution as the first.

Example 13 — Exchangeability: Order Doesn't Matter

Same urn ($r = 3, b = 2$, reinforcement $+1$). Show $\mathbb{P}(R_1B_2) = \mathbb{P}(B_1R_2)$.
Solution.

$$\mathbb{P}(R_1B_2) = \frac{3}{5}\cdot\frac{2}{6} = \frac{6}{30}, \qquad \mathbb{P}(B_1R_2) = \frac{2}{5}\cdot\frac{3}{6} = \frac{6}{30}$$

Equal. In general, $\mathbb{P}$(sequence with k reds, $n-k$ blacks in any specific order) $= \frac{r^{(k)}b^{(n-k)}}{(r+b)^{(n)}}$ (rising factorials), which depends *only on the count* k, not the arrangement.
▶ *Trading parallel:* this is why a Pólya urn is the canonical toy model of **path-independent cumulative reinforcement** — e.g. cumulative order-flow imbalance where only the running totals matter, not the exact sequence of buys/sells that produced them.

Example 14 — The Martingale Property

Let $M_n = \frac{R_n}{R_n + B_n}$ be the red proportion after n draws (urn starts r, b, reinforcement $+1$). Show (M_n) is a martingale: $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n$.
Solution. Given $\mathcal{F}_n$ (so R_n, B_n known, total $T_n = R_n + B_n$), draw $n+1$ is red w.p. M_n, black w.p. $1 - M_n$:

$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n \cdot \frac{R_n+1}{T_n+1} + (1-M_n)\cdot\frac{R_n}{T_n+1}$$

$$= \frac{R_n}{T_n+1}\left[M_n \cdot \frac{T_n}{R_n}\cdot\frac{R_n+1}{R_n}\cdots\right]$$

Cleaner: substitute $M_n = R_n/T_n$ directly,

$$= \frac{R_n}{T_n}\cdot\frac{R_n+1}{T_n+1} + \frac{T_n-R_n}{T_n}\cdot\frac{R_n}{T_n+1} = \frac{R_n}{T_n(T_n+1)}\left[(R_n+1)+(T_n-R_n)\right] = \frac{R_n(T_n$$

⋆ *Why it matters:* by the Martingale Convergence Theorem, $M_n \to M_\infty$ a.s. and in L^1 — the red proportion **stabilizes** to a random limit rather than averaging out to a constant. This is the mathematical seed of path-dependent "lock-in" phenomena (market share, momentum regimes).

Example 15 — The Beta Limit

State (without full measure-theoretic proof) the distribution of $M_\infty = \lim_n M_n$ for a Pólya urn started at (r, b), and verify it's consistent with Example 12.
Solution.

$$M_\infty \sim \text{Beta}(r, b)$$

Check: $\mathbb{E}[M_\infty] = \frac{r}{r+b} = \frac{3}{5}$, matching Example 12's $\mathbb{P}(R_2) = \frac{3}{5}$ exactly (a martingale's expectation never moves: $\mathbb{E}[M_n] = \mathbb{E}[M_0] = \frac{r}{r+b}$ for *every* n, including $n = \infty$).
▶ *Deep link:* this is *the* bridge between discrete reinforcement and continuous Bayesian updating — a Beta(r, b) prior updated by Binomial draws produces exactly this urn's predictive probabilities (Pólya urns are the combinatorial face of Bayesian conjugacy).

Example 16 — Trading Analogy: Momentum via Reinforcement

A simplified order-flow model: start "sentiment urn" at 1 buy-signal, 1 sell-signal token. Each minute, draw a token, reveal its signal, return it plus $c = 4$ more of the same type (strong reinforcement). After the 1st minute draws *buy*, find $\mathbb{P}$(2nd minute also buy).
Solution. After drawing buy and adding 4 more buy tokens: urn is $(1+4)$ buy, 1 sell = 5 buy, 1 sell out of 6.

$$\mathbb{P}(\text{2nd buy} \mid \text{1st buy}) = \frac{5}{6} = \boxed{0.8333}$$

⋆ *Intuition:* strong reinforcement (c large relative to initial counts) is exactly what produces *momentum/herding*: one early signal sharply raises the probability of the same signal repeating. $c = 0$ reduces to i.i.d. coin flips (no momentum); $c \to \infty$ after one draw is near-certainty (full lock-in). Tuning c interpolates between "pure noise" and "pure trend."

E. Capture–Recapture & the Lincoln–Petersen Index

Example 17 — Basic Point Estimate

To estimate the number of active bots N trading an illiquid name, you tag $n_1 = 40$ orders on Monday. On Tuesday you sample $n_2 = 50$ orders and find $m_2 = 8$ were Monday-tagged. Estimate N.
Solution.

$$\hat{N} = \frac{n_1 n_2}{m_2} = \frac{40\cdot 50}{8} = \boxed{250}$$

⋆ *Intuition:* m_2/n_2 estimates the population-wide tagged fraction n_1/N; solving $\frac{m_2}{n_2} \approx \frac{n_1}{N}$ for N gives exactly this formula. It's a method-of-moments estimator in disguise.

Example 18 — Deriving Lincoln–Petersen as a Hypergeometric MLE

Derive $\hat{N}_{\text{MLE}}$ by maximizing the Hypergeometric likelihood $\mathbb{P}(m_2 \mid N) = \frac{\binom{n_1}{m_2}\binom{N-n_1}{n_2-m_2}}{\binom{N}{n_2}}$ over integer N.
Solution. Use the likelihood-ratio trick: form $\frac{L(N)}{L(N-1)}$ and find where it crosses 1.

$$\frac{L(N)}{L(N-1)} = \frac{(N-n_1)(N-n_2)}{N(N-n_1-n_2+m_2)}$$

Setting this ≥ 1 and solving gives $L(N)$ increasing while $N \leq n_1n_2/m_2$ and decreasing after, so

$$\hat{N}_{\text{MLE}} = \left\lfloor \frac{n_1 n_2}{m_2} \right\rfloor = \boxed{\frac{n_1 n_2}{m_2}}$$

confirming Example 17's formula is not just intuitive but exactly the MLE.
▶ This likelihood-ratio "find where it flips from > 1 to < 1" technique is the standard way to MLE a *discrete* parameter — you can't set a derivative to zero, so you difference instead.

Example 19 — Chapman's Bias Correction

Explain why $\hat{N} = n_1n_2/m_2$ is biased upward for small samples, and state the Chapman-corrected estimator. Recompute Example 17 with it.
Solution. $\mathbb{E}[1/m_2] > 1/\mathbb{E}[m_2]$ by Jensen's inequality (Ch. 2.2!) applied to the convex function $1/x$ — so $\mathbb{E}[\hat{N}] > N$ ($\hat{N}$ over-estimates on average), especially when m_2 is small and its distribution right-skewed near 0.
Chapman's estimator:

$$\hat{N}_C = \frac{(n_1+1)(n_2+1)}{m_2+1} - 1$$

Recompute: $\hat{N}_C = \frac{41\cdot 51}{9} - 1 = \frac{2091}{9} - 1 = 232.33 - 1 = \boxed{231.33}$
⋆ *Intuition:* adding 1 to every count is a small-sample "continuity" fix (same philosophy as Laplace smoothing) that specifically tames the $1/m_2$ blow-up risk when m_2 is small or could be 0.

Example 20 — Capstone: Hidden Liquidity, FPC and LP Combined

You want the total number of distinct participants N behind anonymized flow in a dark pool. Monday's 60 distinct tags: $n_1 = 60$. Tuesday you sample $n_2 = 45$ distinct tags and find $m_2 = 15$ overlap with Monday's set. (a) Find $\hat{N}_C$ (Chapman). (b) Treating Tuesday's sample as literally Hypergeometric$(\hat{N}_C, n_1, n_2)$, compute the FPC factor for this sample.
Solution. (a)

$$\hat{N}_C = \frac{61\cdot 46}{16} - 1 = \frac{2806}{16} - 1 = 175.375 - 1 = \boxed{174.375}$$

(b) FPC $= \dfrac{\hat{N}_C - n_2}{\hat{N}_C - 1} = \dfrac{174.375 - 45}{173.375} = \dfrac{129.375}{173.375} = \boxed{0.7463}$

▶ *Full-circle intuition:* Sections A–C gave you the forward problem (known N, K, find the chance of an overlap count); Section E inverts it (observed overlap, estimate the unknown N) — and part (b) shows the FPC isn't just decorative here: with $n_2/\hat{N}_C \approx 26\%$, ignoring it would materially mis-size any confidence interval you build around $\hat{N}_C$ using the Normal approximation of Example 10.

One-tool-tree reminder: Hypergeometric is "Binomial that remembers what it already drew" (negative dependence, shrinking variance via the FPC); Pólya urns run that dependence the other direction (positive reinforcement, growing lock-in, Beta limit); capture-recapture inverts the whole hypergeometric model to estimate an unknown N from an observed overlap. Three faces of the same finite-population coin.